Current dependencies

Default example first / Complete grouped inventory follows / Coach discussion guide

A dependency is something an investment relies on. Its failure can reduce value or block withdrawal. The current app traces one selected PT investment. This document uses the discovery capture for branch **vinkas**, commit **e779c5a**.

Start here: the default example

We lend USDC in the **PT-reUSD-10DEC2026 / USDC** market. Borrowers provide PT-reUSD as security. **PT** means Principal Token: a claim redeemable under its terms at a specified date.

16 graph boxes = our investment + 15 dependency nodes. There are 17 connections. The preset size of $1 million is an input, not an actual fund holding.

Category	Nodes in this example	Why they matter
Lending market (1)	PT-reUSD-10DEC2026 / USDC	Where our USDC is lent.
Assets (3)	PT-reUSD-10DEC2026; reUSD; USDC	Collateral, its discovered backing asset, and the token we lend.
Trading pool (1)	Pendle market reUSD	A pool used to buy or sell the PT.
Pricing contract (1)	Oracle PT-reUSD-10DEC2026	The contract used to value collateral.
Price input (1)	Ojo PT Feed Pendle Chainlink-compatible Oracle	An input read by that oracle.
Organisations (2)	Circle; Issuer of reUSD (unverified)	Registry issuer and an unresolved issuer placeholder.
Control checks (3)	PT-reUSD and reUSD: no owner or proxy detected; USDC: single key (EOA)	Scanner classifications; not a complete permission audit.
Software (2)	Morpho Blue (core); Pendle V2	Runs lending and PT trading.
Blockchain (1)	Ethereum	Processes transactions and records ownership.

Coach focus: which of these can cause a loss or prevent exit? Which should remain visible, which can be grouped, and what is missing? No-owner findings and unverified issuer placeholders need a different visual treatment from confirmed components.

Scope and category counts

Eight separate preset graphs / 61 distinct dependency IDs across their combined inventory

The app offers two investment types: **hold a PT token**, or **lend in a Morpho market backed by PT collateral**. The eight presets are alternatives to analyse. They are not eight confirmed holdings in one portfolio.

Dependency category	Distinct nodes	What it covers
Lending markets	3	Borrowing arrangements against PT collateral.
Tokens and backing	18	PTs, their backing tokens, and USDC.
Pendle trading pools	6	Places to buy or sell a PT.
Market pricing contracts	3	Oracles valuing lending collateral.
Oracle input feeds	2	Individual inputs read by an oracle.
Issuers / backing organisations	9	Named organisations and unresolved placeholders.
Token control checks	17	Detected or unresolved minting / upgrade controls.
Financial software	2	Morpho Blue and Pendle V2.
Blockchain	1	Ethereum.

Preset key used in the following tables

Letters identify where a node appeared; All means A-H. Counts include the investment box; dependency counts exclude it.

Key	Preset investment	Boxes	Deps.	Links
A	PT-reUSD as collateral (USDC supply)	16	15	17
B	PT-USD3 as collateral (USDC supply)	16	15	18
C	PT-sUSDe as collateral (USDC supply)	17	16	19
D	PT-reUSD (held)	9	8	8
E	PT-sUSDe (held)	11	10	11
F	PT-strUSD (held)	12	11	11
G	PT-fxSAVE (held)	12	11	11
H	PT-wstETH (held)	12	11	12

No bridge node appeared in these eight Ethereum traces. Other tokens, chains and bridges exist in the discovery registry, but registry entries are not automatically dependencies of a selected investment.

Scope note: preset H is named PT-wstETH, while its PT node is labelled PT-stETH-30DEC2027. Check the actual contract and redemption terms.

Lending markets and assets

Complete inventory / Categories 1-2 / Preset letters are explained on page 2

01 Lending markets / 3 nodes

In these names, **X / USDC** means borrowers supply X as collateral and borrow USDC. Collateral is security for a loan.

Node name	Role or discovery status	Presets
PT-reUSD-10DEC2026 / USDC	USDC lending market with PT collateral.	A
PT-USD3-17DEC2026 / USDC	USDC lending market with PT collateral.	B
PT-sUSDE-26NOV2026 / USDC	USDC lending market with PT collateral.	C

02 Tokens and their backing / 18 nodes

An **asset** is something held, lent or used as security. A PT can depend on another token that itself depends on further backing.

Node name	Role or discovery status	Presets
PT-reUSD-10DEC2026	Principal Token; exact redemption terms need verification.	A, D
reUSD	Discovered backing token of PT-reUSD; issuer unresolved.	A, D
USDC	Loan and repayment token for the three lending examples.	A, B, C
PT-USD3-17DEC2026	Principal Token; exact redemption terms need verification.	B
USD3	Discovered backing token of PT-USD3; issuer unresolved.	B
PT-sUSDE-26NOV2026	Principal Token; exact redemption terms need verification.	C, E
sUSDe	Discovered backing token of PT-sUSDe; wraps USDe.	C, E
USDe	Further backing token reached through sUSDe.	C, E
PT-strUSD-26NOV2026	Principal Token; exact redemption terms need verification.	F
strUSD	Backing token reached from PT-strUSD.	F
trUSD	Further backing token reached through strUSD.	F
PT-fxSAVE-29OCT2026	Principal Token; exact redemption terms need verification.	G
fxSAVE	Backing token reached from PT-fxSAVE.	G
fxSP	Further backing token reached through fxSAVE.	G
PT-stETH-30DEC2027	Principal Token; exact redemption terms need verification.	H
wstETH	Backing token reached from preset H.	H
stETH	Further staking token reached through wstETH.	H
WETH	Wrapped ETH reached through stETH.	H

Trading and price discovery

Complete inventory / Categories 3-5 / A price warning is not itself a loss estimate

03 Pendle trading pools / 6 nodes

These are pools used to buy or sell PTs. **AMM** means automated market maker: software trading against a pool of assets. Available liquidity affects selling costs.

Node name	Role or discovery status	Presets
Pendle market reUSD	PT trading and exit pool discovered from the Pendle API.	A, D
Pendle market USD3	PT trading and exit pool discovered from the Pendle API.	B
Pendle market sUSDe	PT trading and exit pool discovered from the Pendle API.	C, E
Pendle market strUSD	PT trading and exit pool discovered from the Pendle API.	F
Pendle market fxSAVE	PT trading and exit pool discovered from the Pendle API.	G
Pendle market wstETH	PT trading and exit pool discovered from the Pendle API.	H

04 Market pricing contracts / 3 nodes

An **oracle** is a contract supplying the lending market with a collateral valuation. It may combine feeds and conversion rates.

Node name	Role or discovery status	Presets
Oracle PT-reUSD-10DEC2026	Pricing contract for the corresponding PT lending market.	A
Oracle PT-USD3-17DEC2026	Pricing contract for the corresponding PT lending market.	B
Oracle PT-sUSDE-26NOV2026	Pricing contract for the corresponding PT lending market.	C

05 Oracle input feeds / 2 nodes

A **feed** is an individual input read by an oracle. A label or Chainlink-compatible interface alone does not establish the provider, calculation or market-price accuracy.

Node name	Role or discovery status	Presets
Ojo PT Feed Pendle Chainlink-compatible Oracle	Named input found through contract reads; calculation needs checking.	A
feed 0xDb87...eF8d	Feed address found; descriptive name unresolved.	B

Review before grouping: the default oracle is classified as not tracking the collateral market price using a name-based heuristic. Validate the feed calculation before accepting that conclusion. Consider expanding feeds only when their oracle is selected.

Organisations and infrastructure

Complete inventory / Categories 6-8 / Known names and unknown placeholders

06 Issuers or backing organisations / 9 nodes

An **issuer** supports token creation or redemption; a **custodian** safeguards backing assets. The engine groups these responsibilities under one node type.

Node name	Role or discovery status	Presets
Issuer of reUSD (unverified)	Unverified placeholder; organisation and backing unresolved.	A, D
Circle	Registry label; verify its role for this token.	A, B, C
Issuer of USD3 (unverified)	Unverified placeholder; organisation and backing unresolved.	B
Ethena	Registry label; verify its role for this token.	C, E
Issuer of strUSD (unverified)	Unverified placeholder; organisation and backing unresolved.	F
Issuer of trUSD (unverified)	Unverified placeholder; organisation and backing unresolved.	F
f(x) Protocol (AladdinDAO)	Registry label; verify its role for this token.	G
Issuer of fxSP (unverified)	Unverified placeholder; organisation and backing unresolved.	G
Lido	Registry label; verify its role for this token.	H

07 Financial software / 2 nodes

A **protocol** is software providing a financial service. Smart contracts are the programs implementing it on a blockchain.

Node name	Role or discovery status	Presets
Morpho Blue (core)	Lending software behind the three lending markets.	A, B, C
Pendle V2	Software behind all six discovered PT trading pools.	All

08 Blockchain / 1 node

The network records ownership and processes transactions. An outage can block action.

Node name	Role or discovery status	Presets
Ethereum	Settlement network reached by every preset graph.	All

Evidence distinction: contract reads and protocol APIs identify some components. Organisation labels come from a maintained registry or from inference. The five unverified issuer boxes need evidence before they are presented as established dependencies.

Token control checks

Complete inventory / Category 9 / Scanner output rather than a full permission audit

09 Token control permissions / 17 nodes

These boxes attempt to identify who may create tokens or alter code. They are shown even when the scanner finds no owner or proxy.

Node label returned by the scanner	Presets
PT-reUSD-10DEC2026 minter: no owner or proxy detected	A, D
reUSD minter: no owner or proxy detected	A, D
USDC minter: single key (EOA)	A, B, C
PT-USD3-17DEC2026 minter: no owner or proxy detected	B
USD3 minter: timelock 168h	B
PT-sUSDE-26NOV2026 minter: no owner or proxy detected	C, E
sUSDe minter: timelock 24h	C, E
USDe minter: timelock 24h	C, E
PT-strUSD-26NOV2026 minter: no owner or proxy detected	F
strUSD minter: multisig 3/5	F
trUSD minter: multisig 3/5	F
PT-fxSAVE-29OCT2026 minter: no owner or proxy detected	G
fxSAVE minter: timelock 72h	G
fxSP minter: timelock 72h	G
PT-stETH-30DEC2027 minter: no owner or proxy detected	H
wstETH minter: no owner or proxy detected	H
stETH minter: no owner or proxy detected	H

Single key (EOA): an externally owned account classification. **Multisig 3/5:** the scanner reports three approvals from five signers. **Timelock:** detected actions are subject to a delay.

No owner or proxy detected: the queried methods and storage slots did not reveal those controls. It does not prove that minting is impossible or that every permission was checked.

Decluttering question: should these findings appear as attributes under their token, rather than separate boxes? Keep minting and upgrade powers distinct: the code selects a controller but does not establish a complete map of all roles.

Coach discussion guide

Choose one investment and its most useful failure scenarios before editing the graph

Opening: "We trace dependencies behind a PT investment and estimate whether a failure could cause losses or block an exit. We want to agree which components matter, what is missing, and which details can be collapsed."

Five questions

Use the default example on page 1. Refer to the full node lists only when needed.

1. Which investment should we focus on?
Holding the PT, or lending against it? Ask for a precise market or token address.

2. What are its three most important failure scenarios?
Let the coach prioritise the realistic problems.

3. How would each failure reach our money?
Ask for the path from component failure to loss or blocked withdrawal.

4. Which hidden dependencies are missing, and what can be grouped?
Ask about backing, control permissions, pricing inputs and exit liquidity.

5. Where can we verify and monitor the important links?
Request contract configuration, documentation or API sources, and useful warning signals.

Leave with

Record these outcomes before ending the conversation.

- [ ] One specific investment and its identifier.
- [ ] Three priority failure scenarios.
- [ ] Relevant dependencies and a clear loss / exit-blocking path.
- [ ] Verification and monitoring sources for the important links.
- [ ] Agreement on what to keep, group, correct, add or defer.

Decision / next step

Source: docs/dependency-discovery-snapshot.json, captured 10 October 2026 at 16:42 UTC; engine commit e779c5a. Detailed addresses and evidence are available in docs/dependency-inventory.md. Counts can change with the selected investment or discovery results.